Comments on
Functional Inequalities on the Quantum Boolean Cube with Applications to Quantum Learning Theory

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Analysis on the dyadic group $\{-1,1\}^n$ (i.e. Boolean cube in the present paper) and its Banach-space-valued generalization have important applications in harmonic analysis, probability theory, and functional analysis. This paper is mainly concerned with developing its noncommutative theory. A natural noncommutative counterpart of dyadic group $\{-1,1\}^n$ is the so-called CAR algebra. The analysis on the CAR algebra goes back at least to [1], [CL93] and [LP98] . Another noncommutative counterpart of the dyadic group $\{-1,1\}^n$ is the matrix algebra $\mathbb{M}_{2^n} := \mathbb{M}_2^{\otimes n}$ (i.e. quantum Boolean cube). This paper focuses on the analytic theory on the matrix algebra $\mathbb{M}_{2^n}$ and its applications.

In my preliminary assessment, the main contributions of this paper are as follows:

(1) It is proved that the quantum Ornstein-Uhlenbeck semigroup is holomorphic on $L_p(\mathbb{M}_{2^n})$, $1 < p < \infty$.

In the classical setting, this result was established by Pisier in his remarkable paper [Pi82a]. To state this in detail, let us also recall basic notation. Note that every function $f : \{-1,1\}^n \to X$ admits the Walsh-Fourier expansion

$$f(\varepsilon) = \sum_{A\subseteq[n]} \widehat{f}(A) w_A(\varepsilon), \quad \varepsilon = (\varepsilon_1, \cdots, \varepsilon_n) \in \{-1,1\}^n, \tag{1}$$

where $(X, \|\cdot\|)$ is a Banach space, $[n] = \{1, 2, \cdots n\}$ and $(w_A)_{A\subset[n]}$ Walsh-Fourier system in $L_2(\{-1,1\}^n)$. Define

$$\Delta(w_A) = |A| w_A, \quad A \subset [n].$$

Then the corresponding heat semigroup is defined by

$$e^{-t\Delta} f = \sum_{A\subset[n]} e^{-t|A|} \widehat{f}(A) w_A.$$

Here, $|A|$ denotes the volume of A. Pisier's remarkable result states that the semigroup $(e^{-t\Delta})_t$ is holomorphic on the vector-valued Lebesgue space $L_2(\{-1,1\}^n, X)$ if and only if X is K-convex. Since then, Pisier's theorem has found numerous applications, including in Boolean analysis. For applications related to the subject of the present paper, see for instance [MN14, EI20b, Vol24].

In this paper, the authors established corresponding holomorphic extension theorem for quantum Ornstein-Uhlenbeck semigroups, see Theorem 3.1 in Chapter 3. It is of considerable interest to establish a noncommutative analogue. This point is further illustrated in Chapters 4 and 5 of the present paper. More precisely, this result is used to study the heat smoothing estimates and Bernstein-Markov type inequalities on the quantum boolean

cube $\mathbb{M}_{2^n}$. Although the proof of Theorem 3.1 in Chapter 3 is inspired by Pisier's result, it incorporates several novel ideas, including an equivalent representation of the semigroup. Moreover, to the best of my knowledge, relatively few works have addressed the holomorphicity of noncommutative semigroups. It is also worthwhile to investigate the holomorphicity of this specific noncommutative semigroup.

As mentioned above, Chapters 4 and 5 mainly concern the heat smoothing estimates and Bernstein-Markov type inequalities on the quantum boolean cube $\mathbb{M}_{2^n}$. These two chapters are motivated by [MN14, HMO17, EI20]. To overcome the difficulties arising from noncommutativity, however, the authors employed on recent developments in noncommutative analysis, including representation theory for noncommutative semigroups and Cowling's theorem, see Proposition 4.5 (in Chapter 4) and Theorem 5.8 (in Chapter 5).

(2) Several interesting correlation inequalities are proved in the quantum Boolean cube setting. Recall that Talagrand [Tal96] proved that for monotone increasing functions $f, g : \{-1, 1\}^n \to \{-1, 1\}$, there exists an absolute constant $C > 0$ such that

$$\begin{aligned} \mathrm{Cov}(f, g) := \mathbb{E}(fg) - \mathbb{E}(f)\mathbb{E}(g) \geq C\Psi\Big(M(f, g)\Big), \\ M(f, g) = \sum_{j\in[n]} \|D_j(f)\|_1 \|D_j(g)\|_1, \end{aligned} \tag{2}$$

where $\Psi(t) = t/\log(\frac{e}{t})$, and D_j is the partial differential operator. The authors first introduce a new notion of monotonically increasing element in the quantum Boolean cube $\mathbb{M}_{2^n}$ via the quantum Walsh-Fourier system, and then establish the quantum analogues correlation inequalities of Harris–Kleitman,Talagrand [Tal96], and Keller–Mossel–Sen [KMS14]. Since the quantum Boolean cube has more partial derivatives $((\partial_{j,\ell})_{j\in n,\ell})$ than its classical counterpart, the definition of monotone increasing elements differs between the two settings, see eq. (7.1) and Definition 7.1 in Chapter 7. This gives rise to new phenomena: the quantum Talagrand correlation inequality established in (Theorem 7.10 in Chapter 7) differs substantially from its classical counterpart. As the authors demonstrate in Example 7.6 (in Chapter 7), a straightforward generalization of Talagrand correlation inequality (2) to the quantum setting is not possible. Instead, the authors introduce a new quantity

$$\mathcal{W}_1(T_1, T_2) = \sum_{j=1}^{n} \sum_{\ell=1}^{3} \|\partial_{j,\ell} T_1\|_1 \|\partial_{j,\ell} T_2\|_1$$

(see eq. (7.11) of Chapter 7), which provides a lower bound in the desired correlation inequality. We note that the new quantum Talagrand correlation inequality (Theorem 7.10 in Chapter 7) is established under a so-called similarity assumption on $T_1, T_2 \in \mathbb{M}_{2^n}$, inspired by the work in [KKM16]. Two other correlation inequalities are also proved.

(3) The authors study the quantum Bohnenblust–Hille inequality and quantum learning theory. The study of the Bohnenblust–Hille inequality has a long history. The inequality has also found a broad range of applications in functional analysis and related fields. Recently, the classical Bohnenblust–Hille inequality has also found applications in learning theory [EI22]. Later, Huang, Volberg, and other researchers proved the Bohnenblust–Hille inequality in the quantum setting, where it has also found applications. In the present paper, the authors continue this line of research, and establish an improved quantum Bohnenblust–Hille inequality with the optimal constant $3^{d/2}$ (see Remark 9.3 in Chapter 9). As an application of the quantum Bohnenblust–Hille inequality, the authors study the

L_1-distance of polynomials on the quantum Boolean cube from tail spaces. Finally, in Chapter 10, the authors study quantum learning. As I understand it, the central problem is to approximate a given function accurately by functions of degree less d. The use of so-called quantum decision tree seems interesting, see the proof of Theorem 10.9.

Conclusion This work is closely related to Banach space-valued analysis, which has developed rapidly in recent years (see e.g. [IvHV20], [CEE23], [CEE24]), as well as to certain lines of research in quantum Boolean analysis (see e.g. [VZ24, RWZ24,BGX25]). The present paper was evidently inspired by these related works. However, it is also clear that the paper contains interesting new results and novel proofs. Some of the methods employed draw on recent advances in noncommutative analysis. In my view, this paper makes a worthwhile contribution and deserves a detailed referee report.

My minor suggestions are listed below

1. The hypercontractivity of the heat semigroup has applications to the to Khintchine's inequality (Banach space-valued setting, also named Kahane-Khintchine inequality). It would be better to mention this. For further details, see Pisier's monograph [2].

2. The classical Bohnenblust–Hille inequality appears to be related to Littlewood's 4/3 inequality. It would be helpful to briefly mention this connection.

3. The title "functional inequalities on the quantum Boolean cube" would better reflect the subject matter of the paper.

References

[1] S. Ayupov, S. Ferleger, and F. Sukochev. Isomorphism between the associative and non-associative L_p-spaces of type II_1 hyperfinite factors. Mathematica Scandinavica (1996): 271-285.

[2] G. Pisier, Martingales in Banach Spaces, Cambridge Studies in Advanced Mathematics, 2016.